Data Visualization & Storytelling

Extracting Insights from Iris, Tips, and Flights Datasets

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From Raw Data to Visual Stories

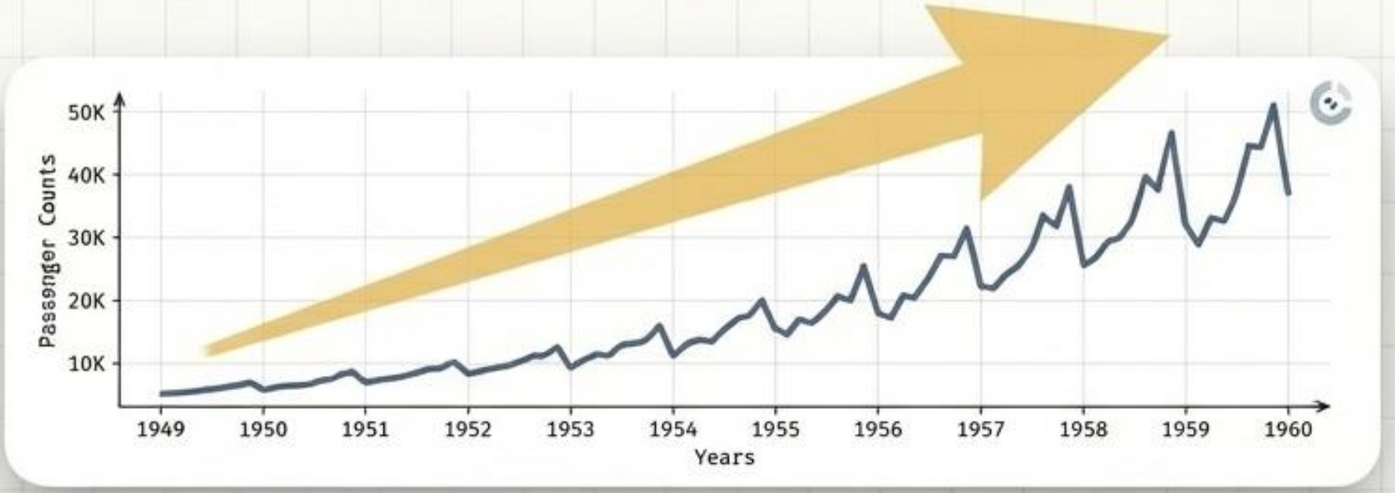
Visualization is not just about making things look pretty; it's about telling a story with data.

In this project, we used `Python` (`Matplotlib` & `Seaborn`) to uncover patterns, distributions, and correlations across three distinct datasets.

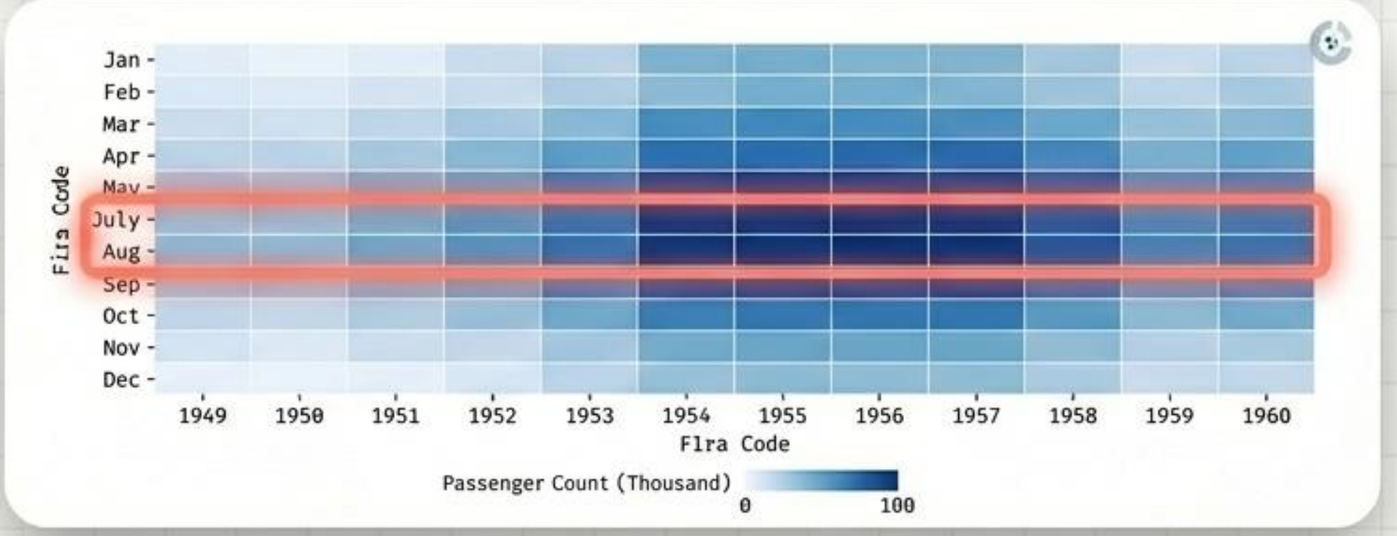
The Golden Age of Aviation

Analyzing monthly passenger counts (1949-1960).

1 1) A clear upward trend showing rapid industry growth year-over-year.



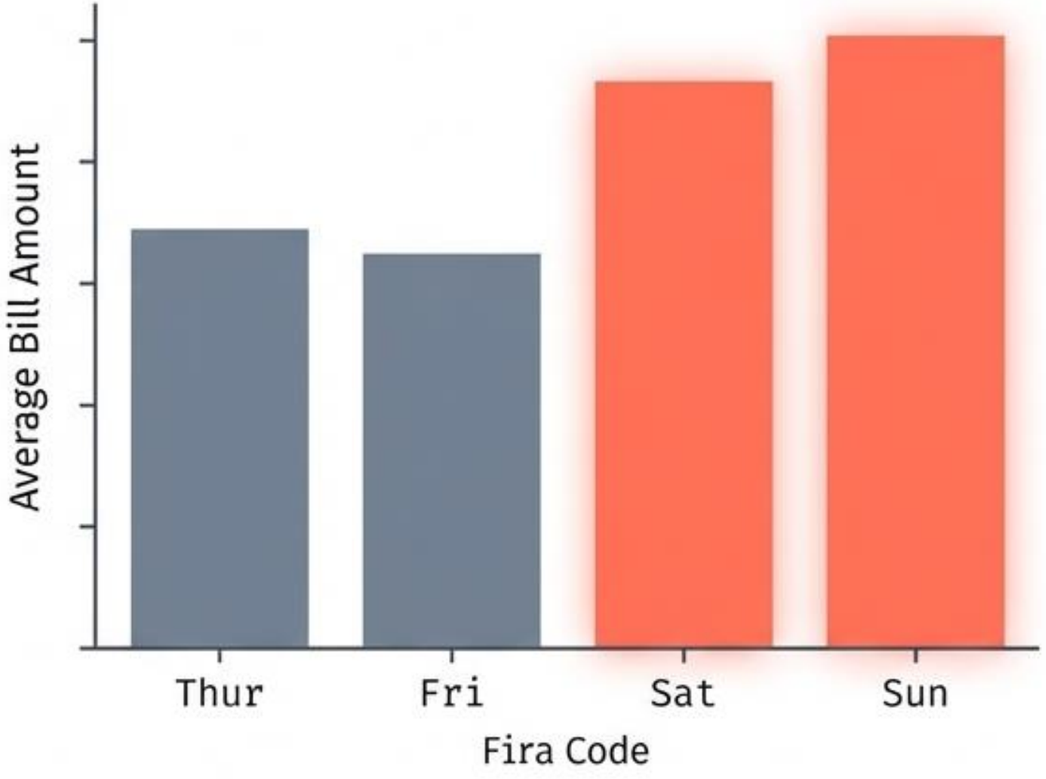
2 2) Strong seasonality, with peak passenger traffic consistently hitting during the summer months (July and August).



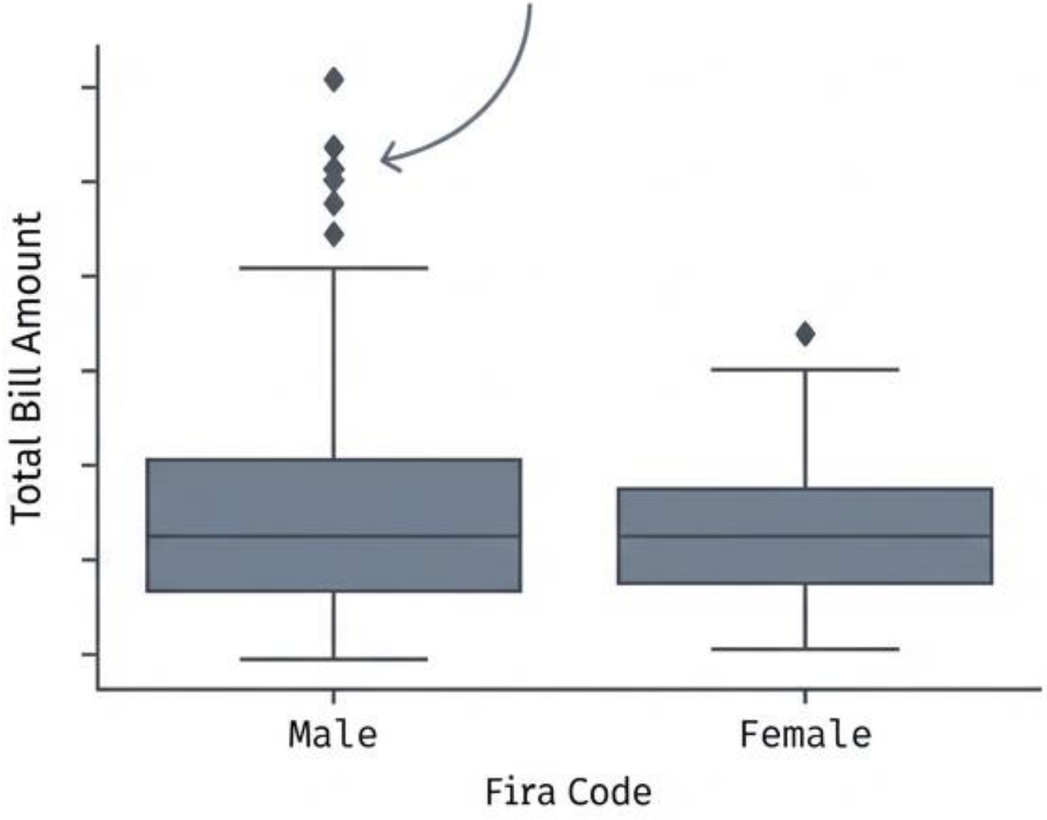
Weekend Spenders

Analyzing restaurant bill and tip data.

1) Customers spend significantly more on average during the weekends (Saturday/Sunday).



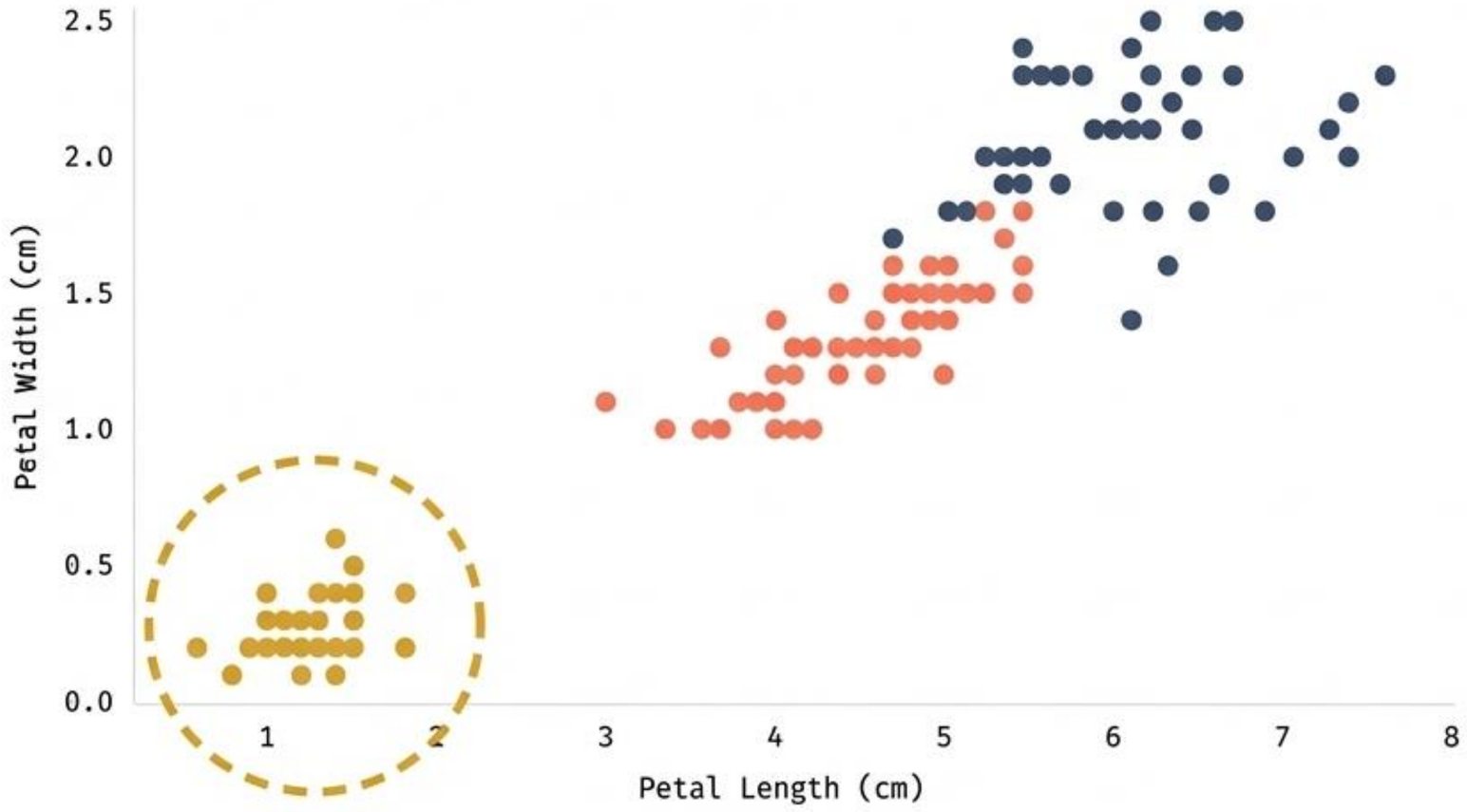
2) Males tend to have a wider range of bill amounts and are responsible for the highest outlier bills.



The Anatomy of Flowers

Analyzing sepal and petal measurements across 3 flower species.

Sepal vs. Petal Dimensions



Correlation Matrix

	Sepal Length	Sepal Width	Petal Length	Petal Width
Sepal Length		0.32	0.58	0.97
Sepal Width	0.19		-0.01	0.23
Petal Length	0.58	0.13		0.96
Petal Width	0.97	0.23	0.96	

Key Insights

1) The 'Setosa' species is highly distinct from others due to its small petal length.

2) There is a nearly perfect positive correlation (0.96) between petal length and petal width.

Choosing the Right Tool

Matplotlib

- The foundational, low-level plotting library.
- Best for highly customized charts where you need exact control over every pixel and layout.

Seaborn

- Built on top of Matplotlib.
- Best for quick statistical analysis, beautiful default themes, and complex charts (like Violins and Heatmaps) with minimal code.

Enter the World of Interactivity

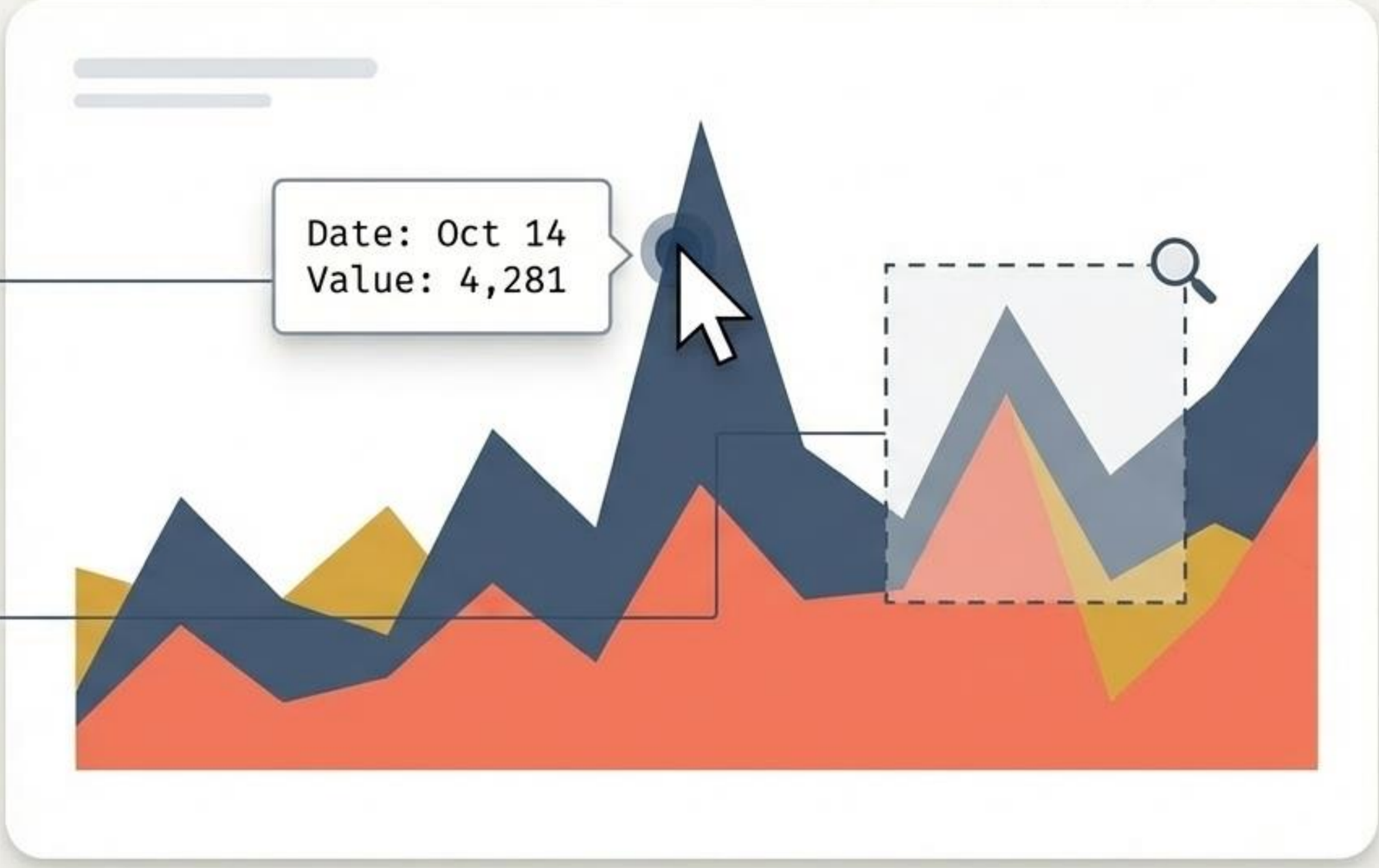
Unlike Matplotlib and Seaborn, which generate static (non-moving) images, Plotly renders graphs in a web browser format (HTML/JavaScript).

1

Key Features: Zoom in, pan around, hover over data points for exact values.

2

Best Use Case: The industry standard for interactive data dashboards and web applications.



Questions & Discussion

Thank you for your time!